Analysis Problem Set #2

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Q1

Defining $z \in \mathbb{R}$ and beginning from the triangle inequality,

$$|x + z| \leq |x| + |z|.$$

Fix z so that $z = y - x$ and equivalently $x + z = y$, and apply algebra and properties of absolute value:

$$|y| \leq |x| + |y - x|$$
$$|y| - |x| \leq |x - y|$$
$$||x| - |y|| \geq |x| - |y| \geq -|x - y|$$

By the properties of absolute value,

$$||y| - |x|| \leq |x - y|.$$

Since x and y are arbitrary, interchanging their labels gives

$$|y| - |x| \leq |y - x| = |x - y| \quad \Longrightarrow \quad -|x - y| \leq |x| - |y|.$$

Taken together,

$$-|x - y| \leq |x| - |y| \leq |x - y|,$$

which is exactly

$$\big||x| - |y|\big| \leq |x - y|. \qquad \blacksquare$$

Q2 (version 1)[1]

For $f(x) = e^x$, $f^{(n)}(x) = e^x$. For $a = 0$, $e^a = 1$.

So the Taylor polynomial ($n = 5$) is:

$$1 + (1)(x) + \tfrac{1}{2}x^2 + \tfrac{1}{6}x^3 + \tfrac{1}{24}x^4 + \tfrac{1}{120}x^5.$$

At $x = 1$, this evaluates to $\approx e$.

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Q2 (version 2)

$T_a(x) = e^a + e^a(x-a) + \frac{e^a}{2!}(x-a)^2 + \frac{e^a}{3!}(x-a)^3 + \frac{e^a}{4!}(x-a)^4 + \frac{e^a}{5!}(x-a)^5$
$T_0(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!}$
$T_0(1) = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!}$
$T_0(1) = \frac{120+120+60+20+5+1}{120}$
$T_0(1) = \frac{326}{120}$

Q3

By the chain rule,

$$\frac{d\log(x(t))}{dt} = \frac{1}{x}\frac{dx(t)}{dt} = \frac{\frac{dx(t)}{dt}}{x}$$

[1] Probably want to choose a version for Q2

Q4

4.1

$$H_f = \begin{bmatrix} -\frac{1}{4}\,x^{-3/2} & 0 \\ 0 & -\frac{1}{4}\,y^{-3/2} \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -\dfrac{1}{x^{3/2}} & 0 \\ 0 & -\dfrac{1}{y^{3/2}} \end{bmatrix}$$

$$|Hf| = \frac{1}{16}\cdot\frac{1}{(xy)^{3/2}} > 0$$

The Hessian is negative definite, so f is strictly concave over $\mathbb{R}^2_{++}$, its domain.

4.2 $g(x,y) = \sqrt{xy}$

$$Hg = \begin{bmatrix} -\frac{\sqrt{y}}{4}\,x^{-3/2} & \dfrac{1}{4\sqrt{xy}} \\ \dfrac{1}{4\sqrt{xy}} & -\frac{\sqrt{x}}{4}\,y^{-3/2} \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -\dfrac{y^{1/2}}{x^{3/2}} & \dfrac{1}{\sqrt{xy}} \\ \dfrac{1}{\sqrt{xy}} & -\dfrac{x^{1/2}}{y^{3/2}} \end{bmatrix}$$

$$|Hg| = \frac{1}{xy} - \frac{1}{xy} = 0$$

The Hessian is negative semidefinite, so g is concave but not strictly concave over $\mathbb{R}^2_{++}$, its domain.

Q5

5.1

For any n, I_n is an open set. Choose any $x \in (-\frac{1}{n}, \frac{1}{n})$. Let $0 < r < \frac{1}{n}$. Then $B(x,r) \subset I_n$.

$$\bigcap_{n=1}^{\infty} I_n = \{0\},$$

because for any $z \in \mathbb{R}$, $z \neq 0$, there exists $n \in \mathbb{N}$ such that

$$d(0, \tfrac{1}{n}) < d(0,z) \iff \left|\tfrac{1}{n}\right| < |z| \implies z \notin I_n.$$

And $0 \in I_n$ because $0 < \frac{1}{n}$ for every $n \in \mathbb{N}$.

$\{0\}$ is a closed set because it is the complement of $(-\infty,0)\cup(0,\infty)$, which is the union of two open balls and thus an open set by way of Theorems 1 and 2.

5.2

For any n, I_n is a closed set. Choose $x = \frac{1}{n+1} \in I_n$. Then for all $r > 0$, $B(x, r)$ is not a subset of J_n

For center $x = \frac{1}{2}$ and radius $r = \dfrac{n-1}{n+1}$,

$$\left(\tfrac{1}{2} - \tfrac{1}{n+1} = \tfrac{n-1}{n+1}\right)$$

we have $\forall y \in J_n$, $d(x, y) \le r$. Also, the complement of J_n is the open set

$$\left(-\infty, \tfrac{1}{n+1}\right) \cup \left(\tfrac{n}{n+1}, \infty\right).$$

Taken together, that implies that J_n is a closed ball.

For $\displaystyle\bigcup_{i=1}^{\infty} J_n$:

$0 < \dfrac{1}{n+1}$ for any n, so for any $u \le 0$, $u \notin \displaystyle\bigcup_{i=1}^{\infty} J_n$.

Similarly $1 > 1 - \dfrac{1}{n+1}$ for any n, so for any $v \ge 1$, $v \notin \displaystyle\bigcup_{i=1}^{\infty} J_n$.

Q6

(1) By definition of homogeneous function of degree one, we have:

$$f(\lambda K, \lambda L) = \lambda f(K, L)$$

We then take derivative of lamda on both side

Q7

Bordered Hessian:

$$H = \begin{bmatrix} 0 & \alpha A K^{\alpha-1} L^{1-\alpha} & (1-\alpha) A K^{\alpha} L^{-\alpha} \\ \alpha A K^{\alpha-1} L^{1-\alpha} & (\alpha^2 - \alpha) A K^{\alpha-2} L^{1-\alpha} & (\alpha - \alpha^2) A K^{\alpha-1} L^{-\alpha} \\ (1-\alpha) A K^{\alpha} L^{-\alpha} & (\alpha - \alpha^2) A K^{\alpha-1} L^{-\alpha} & (\alpha^2 - \alpha) A K^{\alpha} L^{-\alpha-1} \end{bmatrix}$$

where

$$a = \alpha A K^{\alpha-1} L^{1-\alpha}, \qquad b = (1-\alpha) A K^{\alpha} L^{-\alpha}, \qquad c = (\alpha^2 - \alpha) A K^{\alpha-2} L^{1-\alpha},$$
$$d = (\alpha - \alpha^2) A K^{\alpha-1} L^{-\alpha}, \qquad e = (\alpha^2 - \alpha) A K^{\alpha} L^{-\alpha-1}.$$

Leading principal minors

$$|H_2| = -\alpha^2 A^2 K^{2\alpha-2} L^{2-2\alpha} \leq 0 \qquad \text{for } K, L \geq 0$$

$$|H_3| = 2abd - a^2 e - b^2 c$$
$$= 2(\alpha)(1-\alpha)(\alpha-\alpha^2)A^3K^{3\alpha-2}L^{1-3\alpha} - (\alpha^4-\alpha^3)A^3K^{3\alpha-2}L^{1-3\alpha}$$
$$-(1-\alpha)^2(\alpha^2-\alpha)A^3K^{3\alpha-2}L^{1-3\alpha},$$

need to show ≥ 0 for $K, L \geq 0$.

Divide by $(1-\alpha)A^3$:

$$2\alpha^2(1-\alpha)\cancel{K^{3\alpha-2}L^{1-3\alpha}} \ \geq \ -\alpha^3\cancel{K^{3\alpha-2}L^{1-3\alpha}} + (1-\alpha)(\alpha^2-\alpha)\cancel{K^{3\alpha-2}L^{1-3\alpha}}$$

$$\begin{aligned}
2\alpha^2 \ &\geq \ (1-\alpha)(\alpha^2-\alpha) - \alpha^3 \\
2\alpha \ &\geq \ (1-\alpha)(\alpha-1) - \alpha^2 \\
2\alpha \ &\geq \ -\big((\alpha-1)^2 + \alpha^2\big) \\
2\alpha \ &\geq \ -\big(2\alpha^2 - 2\alpha + 1\big) \\
0 \ &\geq \ -2\alpha^2 - 1 \quad \checkmark
\end{aligned}$$

Q8

Q9

1. A is bounded, $M = \sqrt{10}$, $\bar{x} = 0$.
2. B is not bounded. All $x < 0$ are in B.
3. C is not bounded. Assuming $\mathbb{R}_+$ means non-negative, all $(0, y)$ and $(x, 0)$ are in C.
4. D is bounded, $M = 10$ and $\bar{x} = (0, 0)$.